

**BANARAS HINDU UNIVERSITY**

B.Sc. (Hons.) First Semester (Major/Minor) Mid-Semester Test 2024-25

**DEPARTMENT OF COMPUTER SCIENCE****Paper Code: CSCMJ11 / CSCMN11 — Introduction to Programming using C**

Time: 1:00 Hour

Maximum Marks: 20

*Instruction: Answer all questions.***Question 1** *What will be the output of the following codes? Explain your answer with proper reason:* [2 × 4 = 8]

(a) tfamily

```
void main() {  
    float a = 9.0, b = 2.0;  
    int c;  
    c = a / b;  
    printf("%d", c);  
}
```

(b) tfamily

```
void main() {  
    int i = 0, z = 12;  
    if (i == 0 || z > 50)  
        printf("Hello");  
    else  
        printf("Bye");  
}
```

(c) tfamily

```
void main() {  
    int a = 15, b = 13, c = 16, x, y;  
    x = a - 3 % 2 + c * 2 / 4 % 2 + b / 4;  
    y = a / b + 5 - b + 9;  
    printf("x = %d, y = %d", x, y);  
}
```

(d) tfamily

```
void main() {  
    int a = 0, b = 1, c;  
    c = a++ + --b;  
    printf("a = %d, b = %d, c = %d", a, b, c);  
}
```

**Question 2**

[4]

Draw a flowchart for finding the largest number among three numbers  $A$ ,  $B$ , and  $C$ .

---

**Question 3**

[4]

What is the significance of the `break` keyword in a `switch-case` statement? What will happen in the absence of the `break` keyword? Explain with an example.

---

**Question 4**

[4]

Write a C program for printing the sum of squares of the digits of a given integer  $N$ .  
(Example: if  $N = 235$ , then print  $S = 2^2 + 3^2 + 5^2 = 38$ )